

Machine Learning Notes

What is Machine Learning?

ML is a subset of AI that enables systems to learn from data and improve without explicit programming.

Core Concepts

- **Data:** Features (inputs), Labels (outputs), Train/Val/Test splits
- **Model:** Learns parameters to map inputs to outputs
- **Training:** Optimize parameters to minimize error
- **Inference:** Predictions on new data

Types of Machine Learning

1. **Supervised:** Labeled data (Regression & Classification)
2. **Unsupervised:** Unlabeled data (Clustering, Dimensionality Reduction)
3. **Reinforcement:** Reward-based learning

Important Concepts

- Bias-Variance Tradeoff
- Overfitting vs Underfitting
- Cross-Validation
- Evaluation Metrics (MSE, Accuracy, F1, etc.)

Common Algorithms

Linear/Logistic Regression, Decision Trees, Random Forest, SVM, Neural Networks, Deep Learning models (CNN, Transformers)

ML Pipeline

Problem → Data → Preprocessing → Feature Engineering → Training → Evaluation → Deployment

Applications

Recommendation systems, Computer Vision, NLP, Healthcare, Finance, etc.